

Jasen N. Tenney

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Professional Summary

Applied AI full-stack engineer and technical lead with over 20 years of experience building production software and cloud architecture for Fortune 500 companies. Currently leading a six-engineer team that took IFT CoDeveloper, an agentic AI SaaS platform, from concept to over \$1M ARR and 50+ customer tenants by combining LLM workflows, retrieval, and mathematical optimization with React, .NET, Python, and Azure.

Technical Skills

AI & LLM: OpenAI API (Responses API, Structured Outputs), Azure OpenAI, LangGraph, LangSmith, AG-UI, agentic tool calling, evals (promptfoo, LLM-as-judge, golden sets), RAG, vector embeddings, Azure AI Search, prompt engineering, multimodal document extraction, Google OR-Tools (linear programming), Claude Code, MCP

Languages: TypeScript, Python, C#, HTML5, CSS, C++, C

Frameworks: React, Next.js, FastAPI, ASP.NET Core (.NET 10), Marten, Wolverine, Pydantic

Databases: PostgreSQL, MS SQL Server, MongoDB, Pinecone, Azure AI Search

Testing Frameworks: Vitest, Jest, React Testing Library, MSW, Playwright, pytest, xUnit

Tooling & Processes: Azure DevOps, GitHub, git, GitVersion, Semgrep, OWASP ZAP, Agile, Scrum

Cloud Infrastructure: Azure (Container Apps, App Service, Front Door + WAF, Entra External ID, Key Vault, App Configuration, Application Insights, AI Search, PostgreSQL Flexible Server), Docker

Infrastructure as Code: Terraform, Bicep

Experience

Applied AI Full-Stack Software Engineer & Technical Lead, *Improving*, Houston, TX, December 2024 – Present
Client: Institute of Food Technologists (IFT) – CoDeveloper (app.codeveloper.ift.org), an agentic AI platform that helps food scientists formulate and reformulate products

- Technical lead of a six-engineer team that took IFT CoDeveloper from concept to MVP to a production agentic SaaS platform, growing it from \$0 to over \$1M ARR across 50+ customer tenants; top contributor across client, API, formula service, and infrastructure.
- Designed and built an AI formulation engine that reverse-engineers commercial food products from their nutrition labels: OpenAI structured outputs parse ingredient statements, Azure AI Search resolves each ingredient against a USDA branded-foods index, and a Google OR-Tools linear-programming solver computes proportions matching the target nutrition panel.
- Built multimodal document extraction that turns supplier spec-sheet PDFs into structured ingredient records using the OpenAI Responses API with strict JSON schemas, concurrent extraction, conflict-aware merging, provenance tracking, and feature-flagged rollout with per-request telemetry.
- Built the evaluation harness that gates AI quality before release: a golden-set regression suite scoring extraction precision/recall/F1, numeric accuracy, document-type accuracy, schema validity, fabrication checks, p95 latency, and per-model token cost against pass/fail thresholds, plus LLM-as-judge evaluation of workflow output and promptfoo experiments to pick the lowest-cost model that still clears the quality bar.
- Designed and directed the Python agent service behind the product's agentic chat, hands-on across its workflows, skills, and API routes (FastAPI, LangGraph, Pydantic v2, asyncio, uv, pytest): 14 multi-step workflows (reverse engineering, formula optimization, claims and technical-challenge analysis, reasoning chat) composed from 45+ reusable LLM skills with structured outputs, model-tier routing, tool calling with guardrail middleware, and AG-UI streaming.
- Partnered with the project's UX designer, turning her Figma designs into the React 19 / TypeScript client and building it end to end — including the “Sous” agentic chat assistant that renders agent progress, tool calls, and results in real time via AG-UI and Server-Sent Events.
- Architected the multi-tenant ASP.NET Core API (Marten on PostgreSQL, Wolverine messaging) with automated database-per-tenant provisioning, role-based access control, and versioned REST contracts consumed by the client, agent, and formula services.

- Led development of Retrieval-Augmented Generation (RAG) that grounds agent responses in IFT's food science content using chunking, vector embeddings, and Azure AI Search semantic search.
- Architected the Azure platform and directed its Bicep infrastructure as code across four environments: Container Apps, App Service, Front Door with WAF, private networking, PostgreSQL Flexible Server, Key Vault, managed identities, and App Configuration.
- Built Azure DevOps CI/CD for five services (GitVersion versioning, workload identity federation, approval gates, Semgrep and OWASP ZAP scans, Playwright E2E tests), the Docker Compose developer environment, and cross-tier OpenTelemetry observability in Application Insights.
- Led the migration of all users from Azure AD B2C to Microsoft Entra External ID with an idempotent, resumable Microsoft Graph tool, making every service configurable for either provider during cutover.
- Defined team-wide AI-assisted development standards for Claude Code and GitHub Copilot (layered CLAUDE.md guidance, shared settings, spec-driven development), shared through five internal tech talks.

Senior Software Engineer, *Self-Employed*, Houston, TX, November 2022 – December 2024

- Turned Figma designs into responsive React / TypeScript SPAs using the Container/Presentational pattern, Redux Toolkit / RTK Query, React Query, and Zod for maintainable, type-safe code.
- Cut SPA load time 50% by code-splitting routes and lazy-loading components with React lazy.
- Engineered a Node.js / Express REST API for authentication, authorization, and session management, using JWTs in HttpOnly cookies to prevent claims tampering.
- Built Morpheus, an open-source Docker and Kubernetes monitoring product developed under OSLabs, with observability dashboards and AI-powered analysis, using Next.js, Prometheus, Grafana, cAdvisor, Node Exporter, and PostgreSQL.

Technical Director, *Microsoft*, Houston, TX, June 1999 – November 2022

- Migrated on-premises applications to cloud-native Azure architectures for Fortune 500 customers, often re-architecting and rewriting them on App Service, Container Apps, Azure SQL, Key Vault, App Configuration, and Application Insights, with a focus on security, scalability, and observability.
- Built cloud-based .NET web applications from Figma designs in partnership with UI/UX designers, using ASP.NET Core MVC, Razor, Blazor, C# async patterns, Entity Framework Core, REST web APIs, and SQL Server.
- Designed and built a distributed warehousing system (ASP.NET Core, C# mobile apps, .NET web services) that improved operational efficiency for 20,000+ users managing multi-warehouse food distribution.
- Designed a franchise point-of-sale system (C# rich clients, web services, SQL Server, satellite communications) that reduced per-sale transaction times, and a self-updating wellbore visualization app (C#, WPF, SQLite) used for oil and gas well calculations.
- Led delivery of an auditing data warehouse storing millions of rows in SQL Server partitioned tables, and built Power BI / OLAP reporting integrated into Microsoft Teams for daily KPI use.

Education

University of Louisiana at Lafayette, Lafayette, LA

Bachelor of Science in Computer Science, Summa Cum Laude

Awarded Shell Technical Scholarship for Undergraduate Students

Technical Certifications

Anthropic: Building with the Claude API | Claude Code in Action | Introduction to Agent Skills | Introduction to the Model Context Protocol. **Microsoft:** Architecting Azure Solutions | Developing Web Services with .NET | Certified Database Administrator | Power Platform Fundamentals

Presentations

Tech Talks — Improving: *How to Trust AI | Spec Driven Development for Developers | Claude Code Superpowers | The Bleeding Edge of Agent–User Interactivity | AI for Developers*; **Remake Labs:** *Vite and the Modern Frontend DX*